Optimal Time-Stepping with Non-Iterative Defect Correction for Coupled Reaction–Diffusion Equations

Aditya Tiwari

Living document, version 0.3 — September 16, 2026

Repository: `OTS-NIDC` (this paper is generated from it)

1 Review for Single Variable PDEs

Throughout, $\delta_x^2 u_j = (u_{j+1} - 2u_j + u_{j-1})/\Delta x^2$ is the centered second difference and refinement means $\Delta x \to 0$ with Δt slaved to Δx.

1.1 The scalar family and the optimal time step

Consider the deterministic family

$$u_t = D\,u_{xx} + \alpha\,u, \qquad D > 0,\ \alpha \in \mathbb{R}, \tag{1}$$

whose stochastic motivation (the moment equations of (**??**), with $\alpha = \frac{k(k-1)}{2}\sigma^2$) is deferred to §**??**. The pure diffusion case $\alpha = 0$ is exactly Chu's setting (Chu, 2011).

Apply forward Euler in time and the three-point stencil in space. Taylor expansion of both sides, followed by repeated differentiation of (1) to convert time derivatives of u into space derivatives (the operators $D\partial_{xx}$ and α commute, giving the binomial identity $u_{t^k} = (D\partial_{xx} + \alpha)^k u$), yields the local truncation error organized by derivative type:

$$\tau \;=\; \underbrace{\left(\tfrac{D^2\Delta t}{2} - \tfrac{D\Delta x^2}{12}\right)u_{xxxx}}_{\text{Group 1}} \;+\; \underbrace{\alpha D\,\Delta t\,u_{xx}}_{\text{Group 2}} \;+\; \underbrace{\tfrac{\alpha^2\Delta t}{2}\,u}_{\text{Group 3}} \;+\; O(\Delta t^2, \Delta x^4). \tag{2}$$

Only Group 1 couples Δt to Δx. Setting it to zero:

Theorem 1 (Optimal time step (Chu, 2011))**.** *For (1) discretized by forward Euler / centered differences, the unique step size annihilating the leading (u_{xxxx}) truncation term is*

$$\boxed{\Delta t_{\text{opt}} = \frac{\Delta x^2}{6D}} \tag{3}$$

and no single step annihilates both the u_{xxxx} and u_{xxxxxx} truncation terms, since the level-2 condition is $\Delta t = \Delta x^2/(\sqrt{60}\,D) \neq \Delta t_{\text{opt}}$.

(Proof: one-line cancellation in (2); uniqueness by the level-2 computation. Full algebra in `sign-change.tex` §1.)

Proposition 2 (α-independence)**.** *Δt_{opt} is independent of the reaction parameter α: the cancellation operates on the diffusive group alone. For the stochastic hierarchy this makes Δt_{opt} independent of the noise amplitude σ and of the moment order k.*

Following the derivation inspecting the truncation error, at $\Delta t = \Delta t_{\text{opt}}$ the Group-1 term vanishes and the surviving residual is

$$\tau\big|_{\Delta t_{\text{opt}}} = \frac{\alpha \Delta x^2}{6} u_{xx} + \frac{\alpha^2 \Delta x^2}{12D} u + O(\Delta x^4). \tag{4}$$

Then the truncation error τ as assembled in (2) is the *forward* residual: exact − scheme = $\Delta t\, \tau + \cdots$. Cancelling it therefore requires *adding* $\Delta t\, \tau$.

Lemma 3 (Closed form correction term)**.** *The corrected factor at any given time step is,*

$$C_j^{(1)} = +\frac{\alpha\, \Delta x^4}{36D} \delta_x^2 u_j^n + \frac{\alpha^2\, \Delta x^4}{72D^2} u_j^n. \tag{5}$$

Theorem 4 (First-level accuracy)**.** *The corrected scheme*

$$u_j^{n+1} = u_j^n + \Delta t_{\text{opt}} \Big[D\, \delta_x^2 u_j^n + \alpha\, u_j^n \Big] + C_j^{(1)} \tag{6}$$

has step error $O(\Delta x^6)$ and global order 4: the correction $C^{(1)}$ is itself $O(\Delta x^4)$, so its evaluation with the second-order operator δ_x^2 contributes only $O(\Delta x^6)$ to the step — the three-point stencil is mathematically sufficient at $p = 1$.

Status: [V]. Measured orders and errors, with the harness described in Appendix **??**:

Table 1: Scalar $p = 1$ convergence, $u_t = Du_{xx} + \alpha u$ with $D = 0.1$, $\alpha\sigma^2 = 0.25$, Neumann BCs, $T = 0.05$; L^2 error against the exact cosine-series solution. Data: `code/implementation/results.md` (generated by `convergence_test.py`, 2026-07). All methods use the same grid and step size Δt_{opt} at each level.

Method	$N = 32$	$N = 64$	$N = 128$	$N = 256$
OTS-NIDC ($p = 1$)	8.54e-09	5.26e-10	3.27e-11	2.04e-12
Forward Euler	1.23e-05	3.02e-06	7.52e-07	1.88e-07
4th-order FD + FE	1.52e-05	3.79e-06	9.45e-07	2.36e-07
Reference expm	2.77e-05	6.81e-06	1.70e-06	4.23e-07
Measured order (log–log slope, mean over levels):				
OTS-NIDC	4.01	(FE / 4th-order FD / expm: 2.00–2.01)		

The 4th-order-FD row carries the paper's cleanest negative control: high-order spatial accuracy alone buys nothing when the step size is slaved to Δx^2 — the $O(\Delta t)$ time error caps the whole scheme at second order. The boost must come from the time-step structure; that is the method's thesis, verified.

1.2 Stability of Single Variable Systems

phrase all of this as lemmas please god

1.2.1 Setting and criterion

Set $u_j^n = G^n e^{ikj\Delta x}$. The three-point Laplacian has symbol

$$\delta_x^2 u_j^n \;\to\; -\frac{4}{\Delta x^2}\, s\, u_j^n, \qquad s = \sin^2(k\Delta x/2) \in [0,1].$$

Let $\mu = D\Delta t/\Delta x^2$, $\eta = \alpha\Delta x^2/D$ (α already contains σ^2 for the moment equations), and $a = \alpha\Delta t = \mu\eta$. The exact one-step factor is $E(s) = e^{-4\mu s + a}$, so $|E| \le e^a$: for $\alpha > 0$ the equation grows, and $|G| \le 1$ is the wrong requirement. We use the von Neumann condition with growth allowance,

$$|G(s)| \le 1 + C\,\Delta t \quad \text{uniformly in } s \in [0,1], \tag{7}$$

along refinement paths $\Delta t = \mu\Delta x^2/D$ with μ fixed (Lax–Richtmyer). The sharper, natural comparison is $|G(s)| \le e^a$.

1.2.2 Amplification factors

We frame OTS-NIDC as a perturbation of the simple Finite-Euler method, so we begin by deriving the amplification factor G_{FE} and then transform it into G_{NIDC} as follows in,

$$G_{\mathrm{FE}} = 1 - 4\mu s + \mu\eta, \tag{8}$$

$$G_{\mathrm{NIDC}} = 1 - 4\mu s + \mu\eta - \tfrac{\eta}{9}s + \tfrac{\eta^2}{72}, \tag{9}$$

the latter from $C^{(1)}$ in (5): $\frac{\alpha\Delta x^4}{36D}\cdot(-\frac{4s}{\Delta x^2}) = -\frac{\eta}{9}s$ and $\frac{\alpha^2\Delta x^4}{72D^2} = \frac{\eta^2}{72}$. Both are affine in s with negative slope, so the extremes over $s \in [0,1]$ are at $s = 0$ (physical mode) and $s = 1$ (grid mode).

1.2.3 Stability at the optimal step

At $\Delta t = \Delta t_{\mathrm{opt}}$, $\mu = 1/6$, $\eta = 6a$, and

$$G_{\mathrm{NIDC}}(s) = \Big(1 - \tfrac{2}{3}s\Big)(1+a) + \tfrac{a^2}{2}. \tag{10}$$

Compare $E(s) = e^{-2s/3}e^a$: the correction supplies exactly the $a^2/2$ term of e^a at $s = 0$, and $-\frac{2}{3}as$ is the cross term. Since $1 - \frac{2}{3}s \in [\frac{1}{3}, 1]$,

$$\tfrac{1}{3}(1+a) + \tfrac{a^2}{2} \;\le\; G_{\mathrm{NIDC}}(s) \;\le\; 1 + a + \tfrac{a^2}{2} \;\le\; e^a \qquad (a \ge 0).$$

Hence $G_{\mathrm{NIDC}} > 0$ for all s (no oscillatory modes) and $|G_{\mathrm{NIDC}}| \le e^{\alpha\Delta t}$: (7) holds for every $\eta \ge 0$, with no restriction on η. The offset $+\eta^2/72$ is therefore not destabilizing; it is what makes $G(0)$ match e^a to $O(a^2)$. For plain FE at the same μ, $G_{\mathrm{FE}}(s) = (1 - \frac{2}{3}s) + a$, also inside $[\,\frac{1}{3} + a,\, 1 + a\,]$.

1.2.4 Grid-scale boundary for general μ

Requiring the lower side of (7) at $s = 1$, i.e. $G(1) \ge -(1 + \mu\eta)$:

$$\mu_{\mathrm{FE,max}} = \frac{2}{4 - 2\eta} = \frac{1}{2 - \eta}, \tag{11}$$

$$\mu_{\mathrm{NIDC,max}} = \frac{2 - \eta/9 + \eta^2/72}{4 - 2\eta}, \tag{12}$$

with ratio

$$\frac{\mu_{\mathrm{NIDC,max}}}{\mu_{\mathrm{FE,max}}} = 1 - \frac{\eta}{18} + \frac{\eta^2}{144}. \tag{13}$$

(The stricter test $G(1) \geq -1$ gives denominators $4-\eta$ and the same ratio.) At $\eta = 0.30$: $\mu_{\mathrm{FE,max}} = 0.5882$, $\mu_{\mathrm{NIDC,max}} = 0.5788$, ratio 0.98396 ($-1.6\%$). The shrink is $O(\Delta x^2)$, and the effective NIDC CFL is $\Delta t \lesssim (1 - \eta/18)\,\Delta x^2/(2D)$.

1.2.5 Scope of the correction ($\Delta t \neq \Delta t_{\mathrm{opt}}$)

$C^{(1)}$ is derived at $\Delta t = \Delta t_{\mathrm{opt}}$ and is independent of Δt. At $s = 0$,

$$G_{\mathrm{NIDC}}(0) - e^a \;\approx\; \frac{\eta^2}{72}\big(1 - 36\mu^2\big),$$

which is ≤ 0 iff $\mu \geq 1/6$. For $\mu < 1/6$ the excess is Δt-independent, so at fixed Δx with $\Delta t \to 0$ the accumulated growth $(1+\mathrm{const})^{T/\Delta t}$ diverges. Stability statements therefore refer to refinement paths μ fixed, where the excess is $O(\Delta t^2)$ and (7) holds; the $\Delta t/\Delta t_{\mathrm{opt}} \in \{0.5, \ldots, 2\}$ table is a sweep over such paths, not a fixed-Δx study. The method is designed for $\mu = 1/6$.

1.2.6 Pair equations

Replace $D \to D_i + D_j$, $\alpha \to \alpha_{ij}$; (10) holds verbatim with $\mu_{ij} = 1/6$ at $\Delta t_{\mathrm{opt}}^{(ij)}$. For the coupled discrete system there is still no independent proof (open item **??**).

Theorem 5 (Von Neumann stability)**.** *With $s = \sin^2(k\Delta x/2)$, $\mu = D\Delta t/\Delta x^2$, $\eta = \alpha\Delta x^2/D$, the amplification factors are $G_{\mathrm{FE}} = 1 - 4\mu s + \mu\eta$ and*

$$G_{\mathrm{NIDC}} = 1 - 4\mu s + \mu\eta - \tfrac{\eta}{9}s + \tfrac{\eta^2}{72}.$$

For the moment hierarchy $\eta \geq 0$ and the grid-scale stability bound tightens from $\mu \leq 2/(4-\eta)$ to $\mu \leq (2 - \eta/9 + \eta^2/72)/(4-\eta)$: a relative shrink of $1 - \eta/18 + O(\eta^2)$, i.e. $O(\Delta x^2)$, about -1.6% at the coarse $\eta = 0.3$ test point. The optimal step $\mu = 1/6$ sits a factor of 3 inside the CFL ceiling for both schemes.

Status: [V](analytic derivation above + numerical check in `notebooks/ai-src/stability/`; the earlier "expands the CFL" conclusion was an artifact of the pre-correction sign and has been fixed in the same release). The correction is stability-neutral to first order; it never threatens the optimal step. Note the honest asymmetry: this is a *scalar* analysis; the coupled discrete schemes of §2 have no independent stability proof (open item **??**).

1.3 Efficiency profile

The method's cost argument is architectural, not just arithmetic: versus multi-stage Runge–Kutta, OTS-NIDC is single-pass (one read, one write of the state vector; no stage storage, no stage barriers); versus compact schemes, it is strictly local (no tridiagonal solves), so it maps onto GPUs and bandwidth-bound hardware without global communication. At $N = 256$ the verified scalar runs show $\sim 10^3$–$10^4\times$ lower error than forward Euler at identical grid and step [V]— order wins over constants at refinement.

2 The Step to Coupled PDEs

We now couple species to allow for multi-linear systems, a direct expansion from the canon work we expand on!

2.1 Reaction-Diffusion Family and OTS

For N species with diagonal diffusion, linear reaction coupling, and (here) zero noise — the deterministic skeleton of the moment systems to come —

$$\partial_t u^{(i)} = D_i\, \partial_{xx} u^{(i)} + \sum_j R_{ij} u^{(j)}. \tag{14}$$

Lemma 6 (OTS-NIDC for fixed indices)**.** *For equation (14), the optimal step for each $u^{(i)}$ individually, as derived in the single variable case, is exactly $\Delta t^{(i)}_{\text{opt}} = \frac{\Delta x^2}{6D_i}$, with first-level corrections $C^{(1),(i)}$ of the same algebraic form as (5), derived term-by-term in `OTS-NIDC.md` §8–9.*

With this, we get the desired theorem as follows.

Theorem 7 (First-level accuracy for fixed indices)**.** *The scheme*

$$u_k^{(i),n+1} = u_k^{(i),n} + \Delta t_{\text{opt}} \left[D_i \delta_x^2 u_k^{(i),n} + \sum_j R_{ij} u_k^{(j),n} \right] + C^{(1),(i)} \tag{15}$$

has similar step error and global order as given by Theorem 4.

2.1.1 Competing Time Steps and Subcycling

Notice that the scheme presented by Theorem 7 relies on having $u_k^{(j),n}$ from having the value of separate PDE $u^{(j)}$ at time-step n. However, since each equation itself relies on their own time-step, we lose the guarantee of having the required value for any given equation readily available if $D_i \neq D, \forall i$ such that $\Delta t^{(i)}_{\text{opt}} = \Delta t_{\text{opt}}$ for the whole system.

The two failure modes of a naive single step frame the problem.

Failure 1: shared step without correction. If species 1 (fast, D_1) is forced onto species 2's optimum $\Delta t^* = \tau_2$, its Group-1 cancellation breaks. The one-step identity

$$e(\Delta t^*) \;=\; \Delta t^* D_1\, \tilde{u}_{xxxx} \left[\tfrac{\Delta t^* D_1}{2} - \tfrac{\Delta x^2}{12} \right] + O((\Delta t^*)^3) \tag{16}$$

is exact for every Δt^*, and writing $\Delta t^* = \tau_1 + \epsilon$ turns the bracket into $\epsilon D_1/2$ *identically* (no Taylor truncation in ϵ).

We remain, however, that this stencil is not ideal for each step, and it leads to questions of stability. In fact,

Failure 2: naive splitting. Lie–Trotter splitting with the coupling frozen over a macro step H incurs a sustained $O(H)$ forcing error; accumulated over T/H cycles this is a global $O(H) = O(\Delta x^2)$ error — worse than the uncorrected base scheme. Freezing is not viable; coupling data data must be interpolated.

The rest of this paper introduces methods of attempting to corroborate the differences between these time-steps in ways to recover the accuracy that we would lose by just picking one or the other.

3 Regime 1: Explicit Correction Single Stepping

Recall that Eq 16, forcing equation with timestep $\tau_1 \sim D_1$ on Δt^* re-involves the u_{xxxx} term, but writing $h = \tau_1 + \varepsilon$ leads the coefficient of u_{xxxx} to be exactly $\frac{\varepsilon D_1}{2}$. With this, we can formulate a first method of attempting to correspond any two coupled equations.

For any two equations with distinct $\Delta t^{(1)}_{\text{opt}} < \Delta t^{(2)}_{\text{opt}}$, we evolve both equations by $\Delta t^*_{\text{opt}} = \Delta t^{(2)}_{\text{opt}}$ such that $u^(2)$ is optimally solved by nature, and the discretization of $u^{(1)}$ is evolved with $\Delta t^{(1)}_{\text{opt}} + \epsilon$ such that the explicit mismatch correction term recovers order 4 accuracy. Namely,

Lemma 8. *Forcing the time-step h on equation with OTS τ_1, where $h = \tau^{(i)} + \varepsilon$, the **explicit mismatch correction term** is written as,*

$$C^{(i)}_{mismatch} = \frac{hD^2\varepsilon}{2}\delta_x^4 u^{(i),n}$$

where δ_x^4 is evaluated via a five-point stencil and $u^{(i),n}$ is the i-th species evaluated at time n.

Then,

Theorem 9 (OTS-NIDC accuracy fixed indices with time-step forcing)**.** *For equation $u^{(i)}$ with forced time-step h, the scheme*

$$u_k^{(i),n+1} = u_k^{(i),n} + h\left[D_i\delta_x^2 u_k^{(i),n} + \sum_j R_{ij}u_k^{(j),n}\right] + C^{(1),(i)} + C^{(i)}_{mismatch} \tag{17}$$

has similar step error and global order as given by Theorem 4.

3.1 Stability

As we've established that the regime for individual equations remains the same, we prove stability by showing the correction factor described in Lemma 8 is similarly a perturbation on the simple FE scheme. We follow the scalar analysis of §1, now allowing the actual step h to differ from the optimal step $\tau = \frac{\Delta x^2}{6D}$. Write $h = \tau + \varepsilon$ and $r = \frac{h}{\tau}$. We first establish stability of the mismatch-corrected diffusion step, then restore the reaction terms.

Lemma 10 (Mismatch correction and amplification factor)**.** *For $u_t = Du_{xx}$, the corrected update is $u^{n+1} = u^n + hD\delta_x^2 u^n + \frac{hD^2\varepsilon}{2}\delta_x^4 u^n$, where $\delta_x^4 = (\delta_x^2)^2$. With $s = \sin^2\left(\frac{k\Delta x}{2}\right) \in [0,1]$, its amplification factor is $G_{\text{diff}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$.*

Proof. Recall that Lemma 8 that so cancellation requires adding $\frac{hD^2\varepsilon}{2}\delta_x^4 u^n$.

The symbols of δ_x^2 and δ_x^4 are $-\frac{4s}{\Delta x^2}$ and $\frac{16s^2}{\Delta x^4}$. Substituting $h = r\tau$ and $\varepsilon = (r-1)\tau$ gives$G_{\text{diff}}(s) = 1 - \frac{2r}{3}s + \frac{2r(r-1)}{9}s^2$. At $r = 1$, the mismatch vanishes and this reduces to the optimal-step diffusion factor from §1. □

Unlike the reaction corrections in §1, the mismatch contribution need not vanish under refinement at fixed grid frequency: its symbol is $\frac{2r(r-1)}{9}s^2$. We therefore bound the complete factor rather than assume that the correction is uniformly small.

Lemma 11 (Diffusive stability interval)**.** *On an infinite or periodic uniform grid, the mismatch-corrected diffusion step satisfies $|G_{\text{diff}}(s)| \le 1$ for every $s \in [0,1]$ whenever $0 < r \le 4$.*

Proof. The upper bound follows from $G_{\text{diff}}(s) - 1 = \frac{2rs}{9}\left((r-1)s - 3\right) \le 0$, since $0 < r \le 4$ and $0 \le s \le 1$.

For the lower bound, completing the square gives $G_{\text{diff}}(s) = \frac{2}{9}\left(rs - \frac{s+3}{2}\right)^2 + 1 - \frac{(s+3)^2}{18} \ge \frac{1}{9}$. Hence $\frac{1}{9} \le G_{\text{diff}}(s) \le 1$.

For $r > 4$, the grid-scale factor is $G_{\text{diff}}(1) = 1 + \frac{2r(r-4)}{9} > 1$. Thus $r = 4$ is the sharp boundary for this Fourier criterion. □

In particular, the correction preserves diffusive stability throughout the mismatch-corrected range $0 < r < 4$, which extends the uncorrected forward Euler ceiling $r \le 3$, as well as for shortened remainder steps $0 < r < 1$. We retain $r < 4$ as the operating restriction and use it below to recover the reaction-growth bound from §1.

For $u_t = Du_{xx} + \alpha u$, the reaction correction must also be evaluated at the actual step. The same Taylor expansion used in §1 gives $C_{\text{react}}u^n = h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. At $h = \tau$, this is exactly the correction previously derived. Together with the mismatch term, it cancels the leading residual at a general step h.

Theorem 12 (Scalar stability with mismatch correction)**.** *Let $\alpha \ge 0$, and consider $u^{n+1} = u^n + h(D\delta_x^2 u^n + \alpha u^n) + \frac{hD^2(h-\tau)}{2}\delta_x^4 u^n + h^2\alpha D\delta_x^2 u^n + \frac{h^2\alpha^2}{2}u^n$. On an infinite or periodic uniform grid, if $0 < h < 4\tau$, then its amplification factor satisfies $|G(s)| \le e^{\alpha h}$ uniformly in $s \in [0,1]$.*

Consequently, any sequence of admitted steps, including optimal substeps and shortened remainder steps, satisfies $\|u^N\|_{\ell^2} \le e^{\alpha T}\|u^0\|_{\ell^2}$, where $T = \sum_{n=0}^{N-1} h_n$.

Proof. Set $a = \alpha h$ and $r = \frac{h}{\tau}$. Substituting the Fourier symbols gives $G(s) = G_{\text{diff}}(s) + a\left(1 - \frac{2r}{3}s\right) + \frac{a^2}{2}$. The preceding lemma gives $\frac{1}{9} \le G_{\text{diff}}(s) \le 1$. Moreover, for $0 < r < 4$, $-\frac{5}{3} < 1 - \frac{2r}{3}s \le 1$. Therefore, for the upper side, $G(s) \le 1 + a + \frac{a^2}{2} \le e^a$, and for the lower side, $G(s) \ge \frac{1}{9} - \frac{5a}{3} + \frac{a^2}{2} > -(1+a) \ge -e^a$, where the middle inequality is $\frac{10}{9} - \frac{2a}{3} + \frac{a^2}{2} > 0$, which holds for every $a \ge 0$. Hence $|G(s)| \le e^a$.

Over successive steps, the amplification factors multiply. Hence their product has magnitude at most $\prod_{n=0}^{N-1} e^{\alpha h_n} = e^{\alpha T}$. The discrete ℓ^2 estimate follows from Parseval's identity. □

Thus the mismatch correction preserves the scalar growth allowance used in §1 throughout the admitted range $0 < r < 4$. In a subcycle, full steps have $r = 1$ and any shortened remainder has $0 < r < 1$, so both satisfy the same estimate. This establishes the scalar building block; stability of off-diagonal reaction coupling and its interpolation still requires analysis of the complete coupled update.

3.2 Advantage

4 Regime 2: Interpolation and Subcycling

4.1 Setup and switching rule

Two species with $\Delta t_{\text{opt}}^{(1)} < \Delta t_{\text{opt}}^{(2)}$, so $D_1 > D_2$ and species 1 is the fast one. Define the step ratio $r = \frac{\Delta t_{\text{opt}}^{(2)}}{\Delta t_{\text{opt}}^{(1)}} = \frac{D_1}{D_2} > 1$. This is the same $r = h/\tau$ as in Regime 1, with $h = \Delta t_{\text{opt}}^{(2)}$ forced onto the fast equation and $\tau = \Delta t_{\text{opt}}^{(1)}$. The mismatch is $\varepsilon = \Delta t_{\text{opt}}^{(2)} - \Delta t_{\text{opt}}^{(1)} = (r-1)\,\Delta t_{\text{opt}}^{(1)}$, as in Regime 1.

- Macro step $H = \Delta t_{\text{opt}}^{(2)}$. Write $n = \lfloor r \rfloor$, $\theta = r - n \in [0,1)$, and remainder $\rho = H - n\,\Delta t_{\text{opt}}^{(1)} = \theta\,\Delta t_{\text{opt}}^{(1)} \in [0, \Delta t_{\text{opt}}^{(1)})$.

- Switching rule: Regime 1 is stable for $r < 4$ (Theorem 12), so Regime 2 is needed for $r \geq r_c$ with $r_c \in [3, 4)$ a design parameter.
  - $r_c = 3$ is the conservative choice (the uncorrected forward Euler ceiling).
  - Larger r_c is supported by Fourier stability up to $r = 4$, but at $r = 4$ the grid mode is only marginally stable ($G_{\text{diff}}(1) = 1$) and the mismatch term is $O(1)$ at grid scale. Say this explicitly and justify the choice of r_c numerically.
- Ordering within a macro step (slow first):
  1. Advance $u^{(2)}$ from t^k to $t^{k+1} = t^k + H$ with its own optimal step, using $u^{(1)}$ at t^k only (explicit coupling).
  2. Sub-cycle $u^{(1)}$ over $[t^k, t^{k+1}]$, reading $u^{(2)}$ from an interpolant built on the two known levels (and optionally history).

  This makes the slow-species coupling data a true interpolation, never an extrapolation.
- Only the *coupling data* needs interpolating. The fast species' own state is advanced by the Regime 1 scheme.

4.2 How much interpolation is required

Let $I_q u^{(2)}$ denote the Lagrange interpolant in time of degree q through $q+1$ known slow levels, and let p be the level of the correction tower (global order $2p + 2$).

Lemma 13 (Interpolation error)**.** *For $t \in [t^k, t^{k+1}]$ and $u^{(2)}$ sufficiently smooth in time,*

$$\left|u^{(2)}(t) - (I_q u^{(2)})(t)\right| \leq C_q \left(\Delta t^{(2)}_{\text{opt}}\right)^{q+1} \max |\partial_t^{q+1} u^{(2)}|,$$

where C_q is the Lagrange node-polynomial constant. In particular $C_1 = 1/8$.

- Proof: standard Lagrange remainder, no new content. Cite and state the constants for $q = 1, 2$.
- Remark: since $\partial_t^2 u = D_2^2 \partial_x^4 u + \dots$ and $\Delta t^{(2)}_{\text{opt}} = \Delta x^2/(6D_2)$, the linear case gives an error of about $\Delta x^4 |u_{xxxx}|/288$, independent of D_2.

Theorem 14 (Degree of interpolation)**.** *Suppose the fast update uses coupling data $R_{12}(I_q u^{(2)})$. Each sub-step has local error $O(\Delta x^{2q+4})$ from the coupling data, hence the global order contributed is* $\min(2p + 2,\ 2q + 2)$. *In particular, $q \geq p$ is necessary and sufficient to preserve order $2p + 2$.*

- Proof sketch to write out: the interpolation error enters multiplied by the step $h \sim \Delta x^2$, then accumulate over $O(\Delta x^{-2})$ steps. Constants grow with r (the interpolation interval is $\Delta t^{(2)}_{\text{opt}} = r\, \Delta t^{(1)}_{\text{opt}}$) but r is bounded by design.
- Consequence table for the paper:
  - $p = 1$: linear interpolation (2 nodes: t^k, t^{k+1}).
  - $p = 2$: quadratic (3 nodes, adds t^{k-1}).
  - Variant: cubic Hermite dense output using the stored derivatives $D_2 \delta_x^2 u^{(2)} + Ru$ that the slow step already computes. Needs no history, gives higher-order coupling error than $p = 1$ requires.

- **[Open derivation]** The correction $C^{(1),(1)}$ on each substep includes a cross term containing $R_{12}\,\partial_t u^{(2)}$. Decide how $\partial_t u^{(2)}$ is supplied (finite difference of the two slow levels, or the Hermite derivative) and check it does not lower the order.

4.3 Closing the macro step

After n full steps of $\Delta t^{(1)}_{\text{opt}}$, the fast species has advanced $n\,\Delta t^{(1)}_{\text{opt}}$ and still needs $\rho = \theta\,\Delta t^{(1)}_{\text{opt}}$ to reach t^{k+1}.

4.3.1 Method A: remainder step

- Take one final step of size ρ with mismatch parameter $\varepsilon_\rho = \rho - \Delta t^{(1)}_{\text{opt}} = (\theta-1)\Delta t^{(1)}_{\text{opt}} < 0$ (the Regime 1 ε for this step) in Lemma 8, coupling data from $I_q u^{(2)}$ at the beginning of the step.
- Uses a five-point stencil for δ_x^4; needs no solution history.

Theorem 15 (Subcycled scheme with remainder step)**.** *Let $r \geq r_c$, $q \geq p$. The scheme above has global order $2p+2$ (order 4 at $p=1$).*

- Proof outline: Theorem 14 for the coupling data; Theorem 9 for each step (the remainder has step ratio $\rho/\Delta t^{(1)}_{\text{opt}} = \theta < 1$); telescoping over the $n+1$ steps per macro step.
- Stability: scalar building block is Theorem 12, since full steps have step ratio $h/\tau = 1$ and the remainder has $0 < h/\tau < 1$. Coupled stability still open (same status as Regime 1).
- Failure mode to remark on: $\theta \to 0$ gives a vanishingly short remainder. Fine for stability ($G_{\text{diff}} \to 1$), but check that the mismatch term $\varepsilon_\rho \to -\Delta t^{(1)}_{\text{opt}}$ behaves.

4.3.2 Method B: extrapolated point

- Take only the n full optimal steps, with no mismatch correction. Then advance the *state* to t^{k+1} by extrapolating the last $q+1$ fast levels forward by a fraction θ of a step.
- The extrapolation error lands on the state directly rather than being multiplied by h, so the degree requirement is one higher than for interpolation.

Lemma 16 (Extrapolation error)**.** *Let E_q be the degree-q extrapolant through $q+1$ consecutive fast levels spaced $\Delta t^{(1)}_{\text{opt}}$ apart, evaluated at $\theta\,\Delta t^{(1)}_{\text{opt}}$ beyond the last node. Then*

$$\left|u^{(1)} - E_q u^{(1)}\right| \leq \Lambda_q(\theta)\left(\Delta t^{(1)}_{\text{opt}}\right)^{q+1} \max|\partial_t^{q+1} u^{(1)}|, \qquad \Lambda_q(\theta) = \frac{\prod_{i=0}^{q}(\theta+i)}{(q+1)!}.$$

- $\Lambda_1(\theta) = \theta(1+\theta)/2$ and $\Lambda_2(\theta) = \theta(\theta+1)(\theta+2)/6$; both are ≤ 1 for $\theta \in [0,1]$.
- Remark: since $\partial_t^{q+1} u \sim D_1^{q+1}\partial_x^{2q+2} u$ and $\Delta t^{(1)}_{\text{opt}} = \Delta x^2/(6D_1)$, the constant is independent of D_1.
- Accuracy range: the extrapolation distance is $\theta < 1$ by construction, i.e. less than one optimal step beyond the last node. Λ_q grows factorially for longer distances, so this should be stated as the reason the method is limited to a single sub-step of extrapolation.

Theorem 17 (Degree of extrapolation)**.** *The extrapolated point contributes a local error $O(\Delta x^{2q+2})$ per macro step, hence global order $\min(2p+2,\, 2q)$. Order $2p+2$ therefore requires $q \geq p+1$: at $p=1$ this means quadratic extrapolation from three levels, and linear extrapolation caps the scheme at order 2.*

- Proof outline: Lemma 16 gives the local error. Telescoping over $T/H = O(\Delta x^{-2})$ macro steps.
- Remark: if the extrapolated value were only used as coupling data (rather than overwriting the state), the error would again be multiplied by h and the interpolation rule $q \geq p$ would apply. Relevant for an alternative fast-first ordering.

Lemma 18 (Fourier symbol of Method B)**.** *With $G(s)$ the single-step factor at the optimal step, the macro amplification factor for the quadratic extrapolant is*

$$M_B(s;\theta) = G(s)^n \left[\tfrac{(\theta+1)(\theta+2)}{2} - \tfrac{\theta(\theta+2)}{G(s)} + \tfrac{\theta(\theta+1)}{2\,G(s)^2} \right].$$

At the grid mode of the diffusive step, $G = 1/3$ and the bracket equals $1 + 2\theta^2$.

- **[Open]** Show $|M_B(s;\theta)| \leq e^{\alpha H}$ uniformly in $s \in [0,1]$ and $\theta \in [0,1)$ for the admitted n. At the grid mode this reduces to $3^{-n}(1+2\theta^2)$, so check the lower bound on n (i.e. on r_c) this implies.
- Do this numerically first and only then try to prove it.

4.4 Comparison of the two closures

- Table to include: stencil width (δ_x^4 five-point for A; none for B), history storage (none for A; $q+1$ levels of the fast species for B), required degree ($q \geq p$ for A, $q \geq p+1$ for B), stability argument (reduces to Theorem 12 for A; needs the macro-symbol analysis for B), behaviour as $\theta \to 1$.
- Recommendation to draft: A as the default (stability is inherited, no history); B as an alternative that avoids the five-point stencil in the final step but needs more history and a separate stability check.

4.5 Numerical validation

- Mirror Table 1 on a two-species linear system with $D_1/D_2 \in \{5, 10\}$, chosen so that $r > 4$, T and boundary conditions as in the scalar test, error against the exact cosine-series solution.
- Rows to include:
  - A with $q = 1$ (expected order 4).
  - A with $q = 0$, a deliberately under-resolved interpolant (expected order 2): negative control for Theorem 14.
  - B with $q = 1$ (expected order 2) and $q = 2$ (expected order 4): negative control and positive result for Theorem 17.
  - Forward Euler subcycled with the same step structure, as baseline.
- Include non-integer and near-integer r (e.g. $\theta \approx 0.1$ and 0.9) to exercise the remainder.
- Also scan $r \in \{3, 3.5, 3.9\}$ in Regime 1 against Regime 2 to justify r_c.

4.6 Open items

- Coupled (off-diagonal) stability for the subcycled-and-interpolated update; only the scalar building block is covered so far.
- Cross-term correction with $\partial_t u^{(2)}$ inside the sub-steps (see above).
- More than two species: define the ordering of the ratios and whether to nest the subcycles.
- $p \geq 2$: interpolation degree $q \geq p$ and extrapolation degree $q \geq p + 1$ are predicted by the same counting, but untested.

A The p-level tower

The general correction is generated by powers of the operator, with the sign of Remark **??**:

$$C_j^{(\le p)} \;=\; +\sum_{\ell=1}^{p} \frac{\Delta t_{\mathrm{opt}}^{\ell}}{(\ell+1)!}\big(D\mathcal{D}_2+\alpha\big)^{\ell} u_j^n, \tag{18}$$

where $\mathcal{D}_2$ is a spatial operator whose accuracy must be raised as ℓ grows. Spatial discretization error breaks the binomial symmetry of the time expansion, giving the stencil-scaling bound:

Proposition 19 (Stencil bound; sketch)**.** *Global order $2p+2$ requires corrections evaluated on at least a $(2p+3)$-point stencil: $p=1$ needs the 3-point main stencil plus a 5-point evaluation inside $C^{(1)}$ when full fourth-order spatial accuracy is demanded of the correction itself; $p=2$ needs a 7-point stencil to represent the uncancelled u_{xxxxxx} residual.*

Status: [O]for $p \ge 2$ numerics (no harness yet; the bound's derivation is a consistency argument in `OTS-NIDC.md` §6). The $p=1$ row is [V]via Table 1.

References

Kevin T. Chu. Boosting the accuracy of finite difference schemes via optimal time step selection and non-iterative defect correction. *Applied Mathematics and Computation*, 218(7):3596–3614, 2011. doi: 10.1016/j.amc.2011.08.108. arXiv:0811.2847.